Policy Learning with Weak Signals

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CODE@MIT November 13, 2026

Motivation

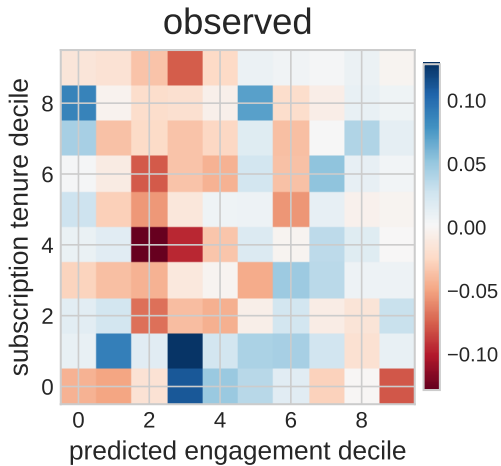
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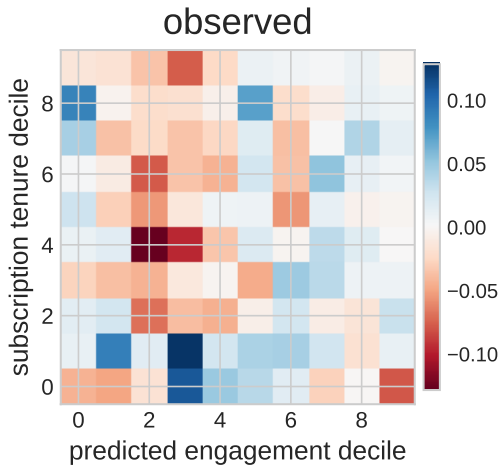
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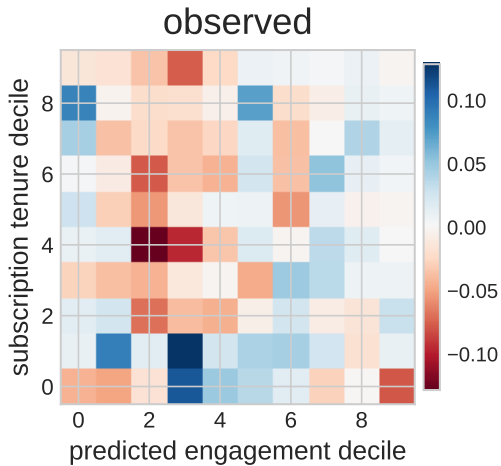
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- Most cells not significant
- What to do?



Our Contribution

In this **weak-signal** regime

- Optimal treatment policy is not learnable.
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However, if effects are **smooth**

- Linear smoothing is both consistent and minimax-optimal.

Decision Problems

For each cell $x_k \in \mathcal{X} = \{x_1, \ldots, x_K\} \subseteq \mathbb{R}^d$

- CATE: $\tau_k = \mathbb{E}[Y(1) - Y(0) \mid X = x_k]$.
- Population share: $p_k = \mathbb{P}(X = x_k)$.
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The **oracle policy** $\pi_k^* = \mathbf{1}\{\tau_k > 0\}$ attains

$$V_{\boldsymbol{\tau}}^* = \sup_{\pi \in [0,1]^K} V_{\boldsymbol{\tau}}(\pi).$$

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  and minimize **regret**

$$V^*_{\boldsymbol{\tau}} - W_{\boldsymbol{\tau}}(\widehat{\pi}) \geq 0.$$

Weak-Signal Regime I

Assumption

For $k = 1, \ldots, K$,

$$\widehat{\tau}_k \overset{\text{ind}}{\sim} \mathcal{N}\left(\tau_k, \frac{\sigma_k^2}{n}\right)$$

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$$\frac{|\tau_k|}{\sigma_k/\sqrt{n}} = \frac{|h_k|}{\sigma_k} = O(1)$$

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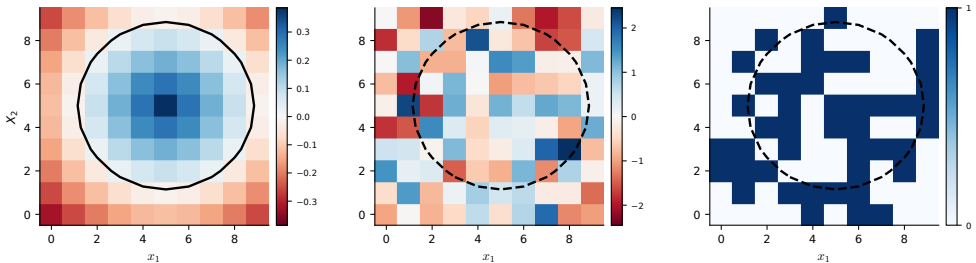
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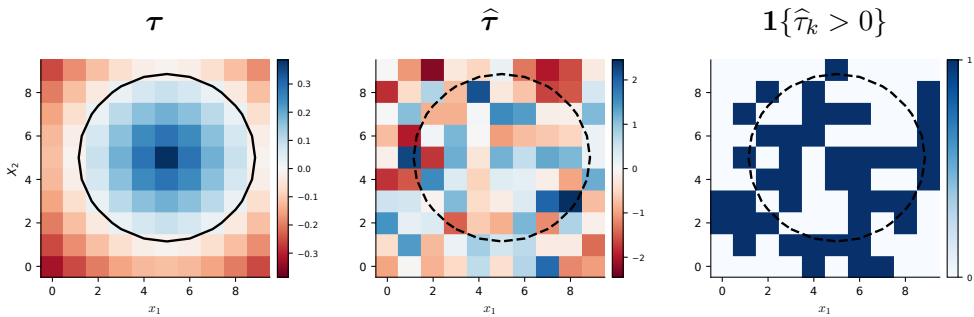
- Each cell remains noisy.
- $K \to \infty$, with n large but fixed: $N = Kn$ grows.

Weak-Signal Regime II

$\boldsymbol{\tau}$ $\widehat{\boldsymbol{\tau}}$ $\mathbf{1}\{\widehat{\tau}_k > 0\}$

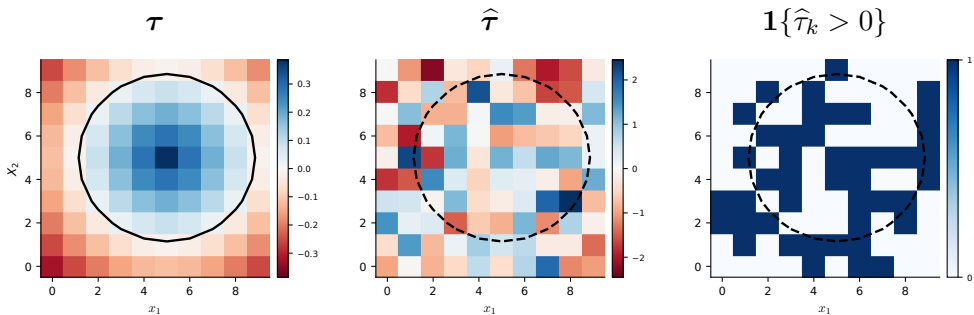


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- Attains only **14%** of the optimal value $V^*_{\boldsymbol{\tau}}$.

An Optimal Policy Is Not Learnable

Theorem 1 (Unattainable Optimal Policy)

$$\inf_{\widehat{\pi}:\mathbb{R}^K \to [0,1]^K} \sup_{\boldsymbol{h} \in [-H,H]^K} \sqrt{n}\left\{V^*_{\boldsymbol{h}} - W_{\boldsymbol{h}}(\widehat{\pi})\right\} = \max_{0 \le t \le H} t\Phi\left(-\frac{t}{\overline{\sigma}}\right) > 0.$$

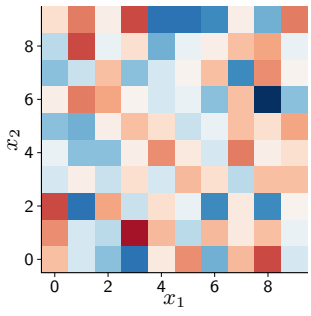
Even the best decision rule has nonvanishing worst-case regret.

- Regret does not vanish as $K \to \infty$.
- Yet we show that $V^*_{\boldsymbol{\tau}}$ is estimable, albeit slowly.

Smooth Signal I

So far, treatment effects are unrestricted across cells.

τ: unrestricted

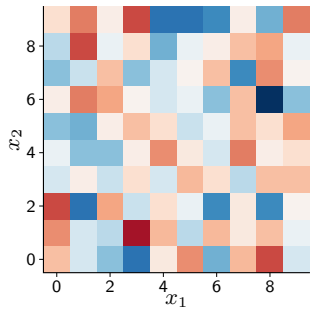


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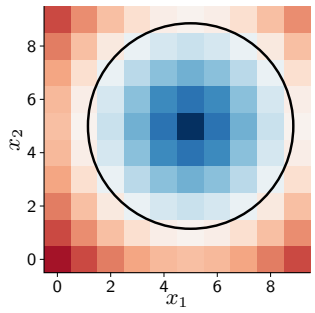
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What if they are smooth?

τ: unrestricted



τ: smooth



Smooth Signal II

Write $\tau_k = g(x_k)$.

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Assumption (Hölder-Smooth Signal)

For fixed $\beta \in (0, 1]$ and $L > 0$,

$$|g(x) - g(y)| \leq L\|x - y\|_2^\beta$$

for all $x, y \in \mathbb{R}^d$.

Nearby cells have similar effects, so we can pool information.

Pooling CATE Estimates

Definition 2 (Linear Smoother at Scale r)

For $r > 0$, pool estimates using deterministic weights:

$$\widehat{\tau}_r(x) = \sum_{j=1}^{K} w_{r,j}(x)\widehat{\tau}_j, \qquad \sum_{j=1}^{K} w_{r,j}(x) = 1.$$

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Treat when the pooled estimate is positive: $\widehat{\pi}_{r,k}(\widehat{\boldsymbol{\tau}}) = \mathbf{1}\{\widehat{\tau}_r(x_k) > 0\}$.

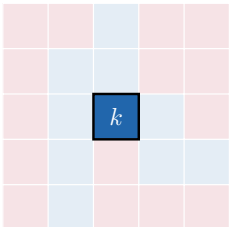
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Target cell: $\tau_k = +0.02$.

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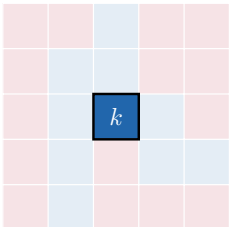
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High estimation error

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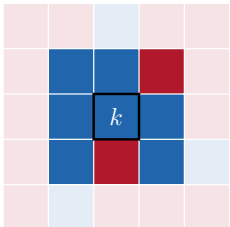
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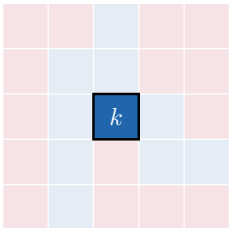
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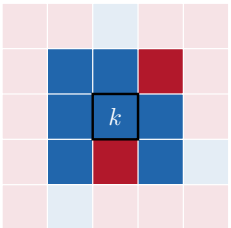
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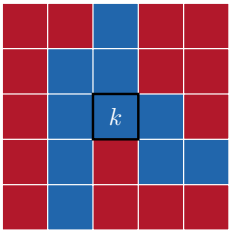
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$\widehat{\tau}_r(x_k) = +0.02$

Local pooling

25 cells



$\widehat{\tau}_r(x_k) = -0.01$

High approximation error

Minimax Optimality of Linear Smoothers

Theorem (Regret Bounds, Informal)

Under regularity conditions,

$$\sqrt{n}\left\{V_{\boldsymbol{\tau}}^{*} - W_{\boldsymbol{\tau}}(\widehat{\pi}_r)\right\} \lesssim \underbrace{Lr^{\beta}}_{\text{approximation error}} + \underbrace{\frac{\overline{\sigma}}{\sqrt{Kr^{d}}}}_{\text{estimation error}} .$$

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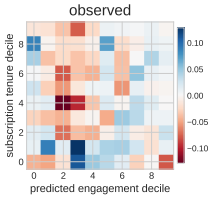
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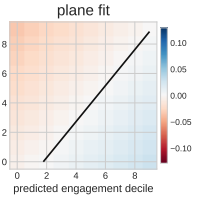
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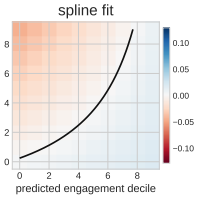
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- We propose an adaptive cross-validation procedure to select r_K^*.
- A matching lower bound establishes **minimax optimality**.

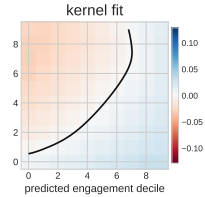
Large Scale Experiment

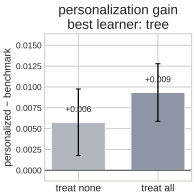


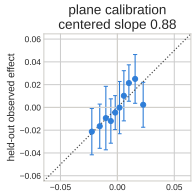


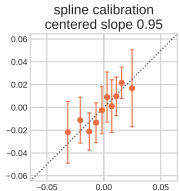


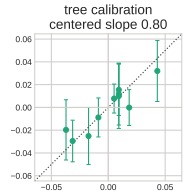


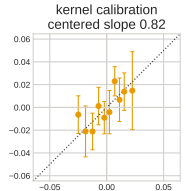










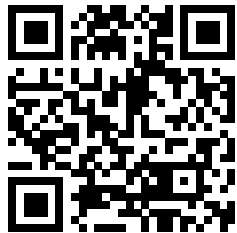


Summary

In the weak-signal regime:

- Without structure, the optimal policy is not learnable.
- The optimal value is estimable, but only slowly.
- With smooth effects, linear smoothing is minimax optimal.
- Optimal resolution balances approximation and estimation error.

Paper



arXiv:2610.10167

Thank you!